

ACDC

DC Circuit Analysis

Direct Current (DC) vs Alternating Current (AC)

- **DC:** Electric charge flows in a single, constant direction. Voltage remains steady over time.
 - **AC:** Electric charge periodically reverses direction, forming a sinusoidal voltage waveform.
-

Voltage, Current, and Power

Voltage (V)

Voltage between two points is the difference in electrical potential energy per unit charge:

$$V = \frac{W}{Q}$$

where:

- W is work (Joules)
- Q is charge (Coulombs)

Current (I)

Current is the rate of charge flow:

$$I = \frac{dQ}{dt}$$

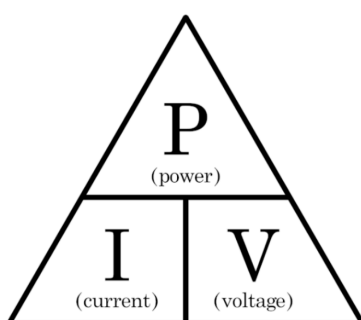
where:

- I is current (Amperes)
- Q is charge (Coulombs)

Power (P)

Power is the rate at which electrical energy is transferred:

Power Formulas



$$P = I \times V$$



$$I = P \div V$$



$$V = P \div I$$

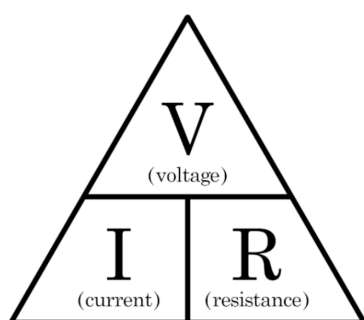
where:

- V is voltage (Volts)
 - I is current (Amperes)
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Ideal Components and Circuit Schematic

Ohm's Law

Ohm's Law Triangle



$$V = I \times R$$



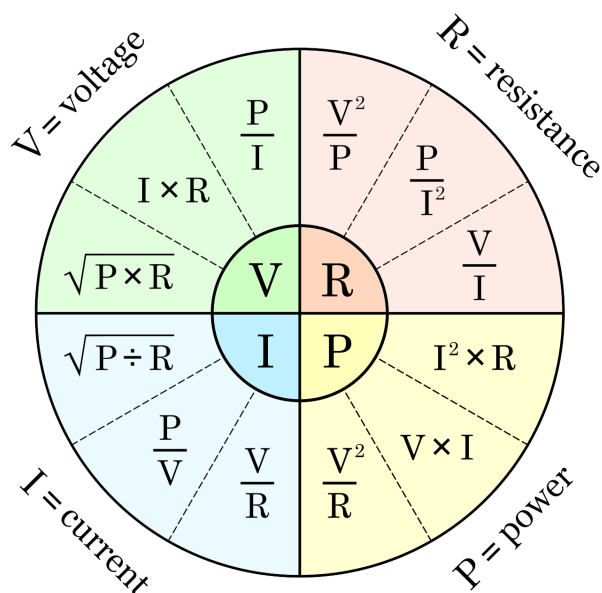
$$I = V \div R$$



$$R = V \div I$$

where:

- V is voltage (Volts)
- I is current (Amperes)
- R is resistance (Ohms, Ω)



Resistance

$$R = \rho \frac{l}{A}$$

where:

- ρ is the resistivity coefficient (Ωm)
- l is length (m)
- A is cross sectional area (m^2)

Circuit Analysis Techniques

Kirchhoff's Voltage Law (KVL)

Sum of voltage drops must equal sum of voltage rises in a closed loop:

$$\sum V_{\text{drop}} = \sum V_{\text{rise}}$$

Kirchhoff's Current Law (KCL)

Sum of currents entering a node is equal to the sum of currents leaving that node:

$$\sum I_{\text{in}} = \sum I_{\text{out}}$$

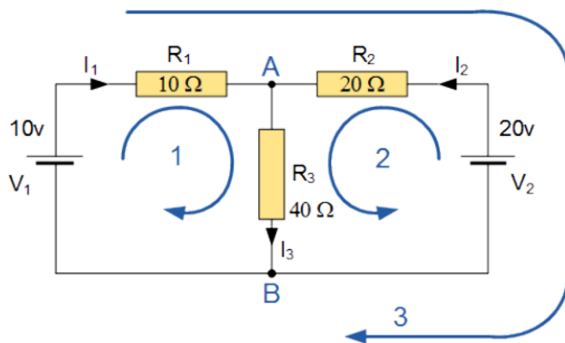
Node Voltage Analysis

Define node voltage using a reference node (0V). Example circuit equations:

$$\frac{V_1 - V_2}{R_1} + \frac{V_2}{R_2} = 0$$

Example: KCL/KVL

Find the current flowing in R_3



Solution

Apply KCL at node A:

$$I_1 + I_2 = I_3 \quad (4)$$

Apply KVL to Loop 1 and 2:

$$10 - I_1 R_1 - I_3 R_3 = 0 \quad (5)$$

$$20 - I_2 R_2 - I_3 R_3 = 0 \quad (6)$$

Substitute (4) into (5) & (6):

$$10 - I_1 R_1 - (I_1 + I_2) R_3 = 0 \quad (7)$$

$$20 - I_2 R_2 - (I_1 + I_2) R_3 = 0 \quad (8)$$

As we know R_1 , R_2 , and R_3 , with two equations, (7) & (8), we can solve for two unknowns, I_1 and I_2 :

$$I_1 = -0.143 \text{ A}$$

Negative; direction of current flow initially chosen was wrong

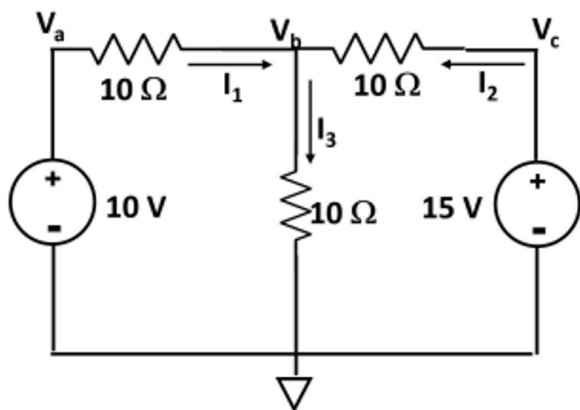
$$I_2 = 0.429 \text{ A}$$

And by (4):

$$I_3 = 0.286 \text{ A}$$

Example: Node Voltage Analysis

Find the currents I_1 , I_2 and I_3 in the circuit below using node voltage analysis



Solution

1. Choose the conductor at the bottom as a reference node.
2. Label all the node voltages and determine the number of unknown ones.
 - $V_a = 10V$ and $V_c = 15V$
 - V_b is unknown.
3. Apply KCL at each unknown node.

$$I_1 + I_2 = I_3$$

4. Express the currents in the KCL equations in terms of node voltages.

$$\frac{V_a - V_b}{10} + \frac{V_c - V_b}{10} = \frac{V_b - 0}{10}$$

$$\frac{10 - V_b}{10} + \frac{15 - V_b}{10} = \frac{V_b - 0}{10}$$

5. Solve the linear equation.

$$V_b = \frac{25}{3}V$$

6. Branch currents can be found from the node voltages obtained.

$$I_1 = \frac{10 - V_b}{10} = \frac{10 - \frac{25}{3}}{10} = \frac{1}{6}A$$

$$I_2 = \frac{15 - V_b}{10} = \frac{15 - \frac{25}{3}}{10} = \frac{2}{3}A$$

$$I_3 = \frac{V_b - 0}{10} = \frac{\frac{25}{3}}{10} = \frac{5}{6}A$$

Thevenin Equivalent Circuits

Simplifies a linear circuit into a single voltage source and a series resistance.

$$V_{th} = \frac{R_2}{R_1 + R_2} V_s$$

Capacitors and Inductors

Capacitor

- Consists of 2 parallel conductive plates separated by a dielectric.

- Blocks DC, allows AC to flow.
- Capacitance (C) is the ability to store charge:

$$C = \frac{Q}{V}$$

- C : capacitance (F), Q : charge (C), V : voltage (V)

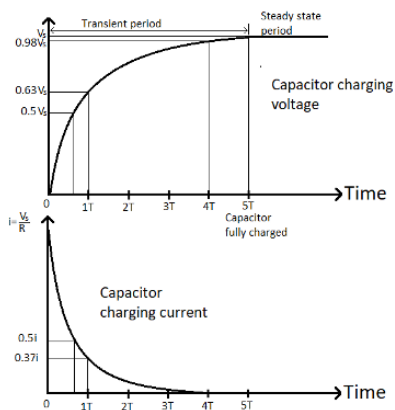
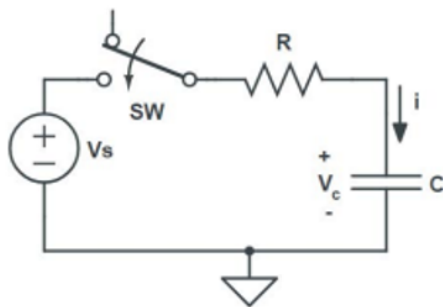
Charging

- At $t = 0$: $V_C = 0$, $I_C = \frac{V_s}{R}$ (Short Circuit)
- Voltage rises, current decreases exponentially.
- Fully charged ($\sim 5\tau$): $V_C = V_s$, $I_C = 0$ (Open Circuit)

$$V_C(t) = V_s(1 - e^{-t/\tau})$$

$$I_C(t) = \frac{V_s}{R}e^{-t/\tau}$$

- $\tau = RC$: time constant.



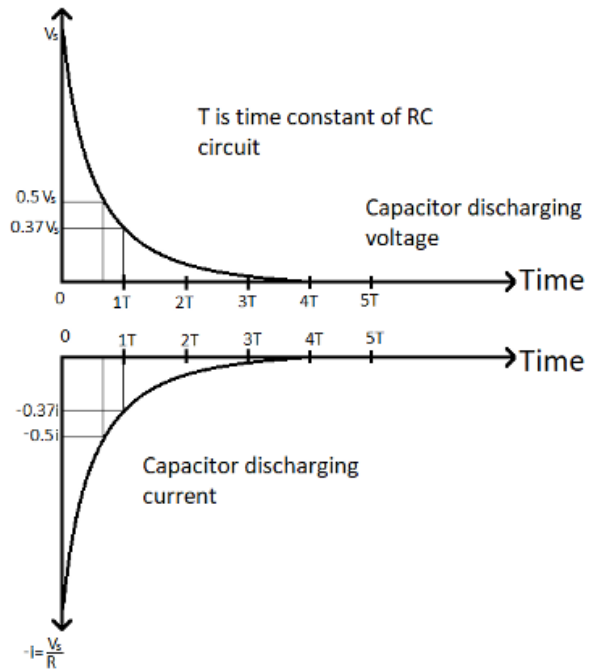
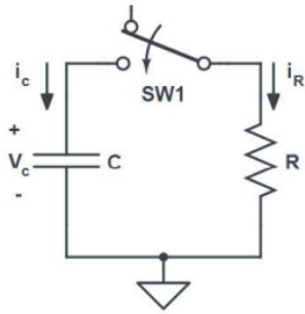
Discharging

- At $t = 0$: $V_C = V_s$, $I_C = \frac{V_s}{R}$
- Voltage and current decrease exponentially.
- Fully discharged ($\sim 5\tau$): $V_C, I_C \approx 0$

$$V_C(t) = V_s e^{-t/\tau}$$

$$I_C(t) = \frac{V_s}{R} e^{-t/\tau}$$

- Discharge current flows opposite to charging current.



Inductors

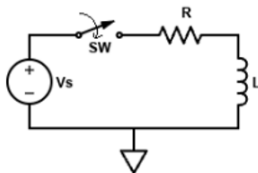
- A conductor wound into a coil.
- Inductance opposes changes in current.

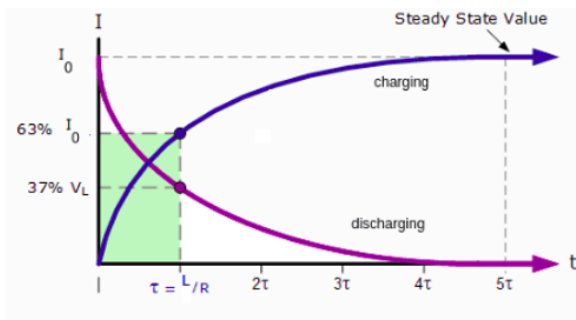
Charging

- At $t = 0$: $I_L = 0$, $V_L = V_s$ (Open Circuit)
- Current rises, voltage decreases exponentially.
- Fully charged ($\sim 5\tau$): $I_L = \frac{V_s}{R}$, $V_L = 0$ (Short Circuit)

$$I_L(t) = \frac{V_s}{R}(1 - e^{-t/\tau})$$

- $\tau = L/R$: time constant.

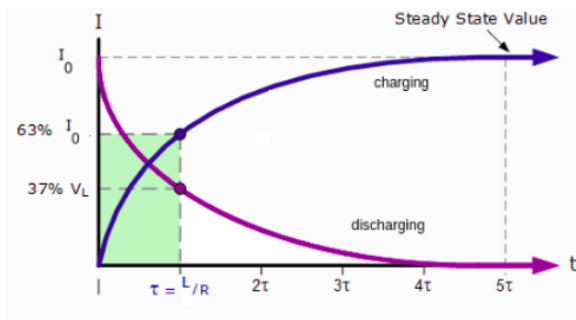
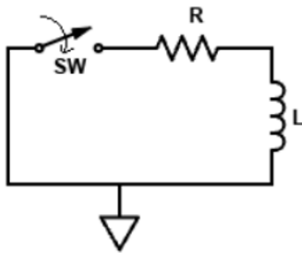




Discharging

- When $V_s = 0$, I_L decreases exponentially:

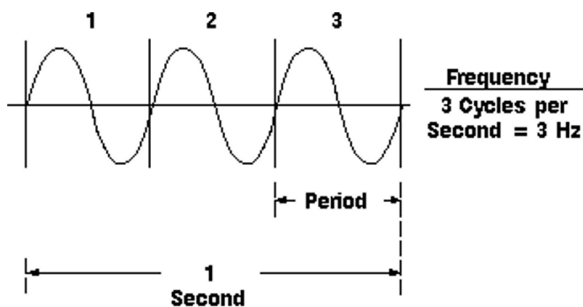
$$I_L(t) = \frac{V_s}{R} e^{-t/\tau}$$



Waveform

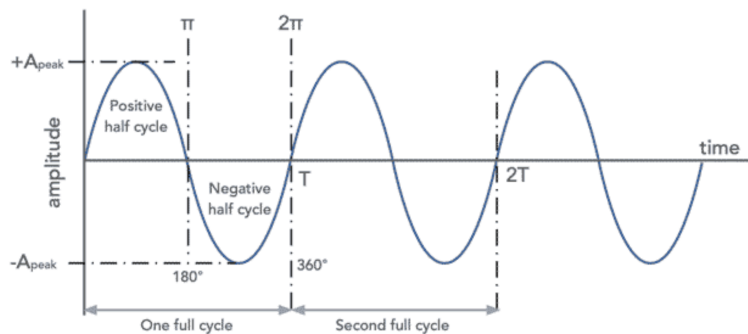
- Period (T) is the time for one complete cycle of a sine wave (in seconds).
- Frequency (f) is the number of cycles per second, measured in Hertz (Hz).

$$T = \frac{1}{f} \quad \text{and} \quad f = \frac{1}{T}$$



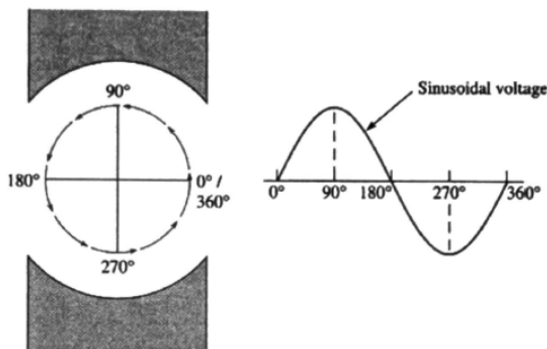
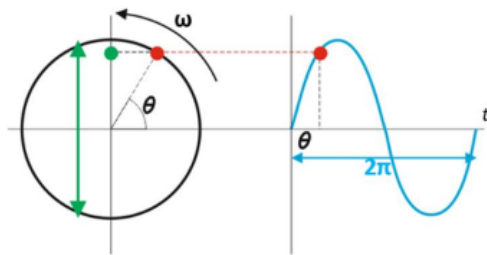
Angular

Periodic waveform, such as $\sin(x)$ or $\cos(x)$, repeats itself every 2π or 360°



Degree: $1/360$ of a complete revolution $Degrees = \frac{180^\circ}{\pi} \times radians$

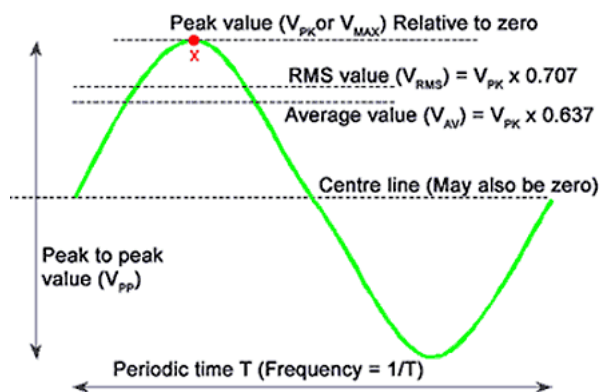
Radian (rad): Measurement along the circumference of a circle which is equal to the radius of the circle $Radians = \frac{\pi}{180} \times degrees$



$$\frac{2\pi}{\omega} = T \text{ (Period)}, \quad \omega = \frac{2\pi}{T}$$

Magnitude

Magnitude of a sine wave changes at every instant as the wave goes through a complete cycle.



1. Instantaneous Value: Magnitude at any particular instant

$$v = V_p \sin(2\pi ft) \text{ V} \quad \text{or} \quad v = V_p \sin \theta \text{ V}$$

$$i = I_p \sin(2\pi ft) \text{ A} \quad \text{or} \quad i = I_p \sin \theta \text{ A}$$

2. Peak Value: + or - maximum value w.r.t to θ reference level
3. Peak-to-Peak Value: Measured from + peak to - peak

$$V_{pp} = 2 \times V_p, \quad I_{pp} = 2 \times I_p$$

4. **RMS Value:** The Root Mean Square (RMS) value represents the effective or equivalent DC value of an AC waveform, meaning it delivers the same power as a DC source of the same magnitude.

$$V_{\text{rms}} = \frac{V_p}{\sqrt{2}}, \quad I_{\text{rms}} = \frac{I_p}{\sqrt{2}}$$

$$V_p = \sqrt{2} \times V_{\text{rms}} \quad I_p = \sqrt{2} \times I_{\text{rms}}$$

5. **Average Value:** The average value is mathematically 0, as the + cancels the - value. However, in many applications, a **rectified** version of the sine wave is used. In a **half-wave rectified** signal, only the positive half-cycle is considered. This value represents the **DC equivalent voltage or current** that would produce the same effect in a resistive circuit. The average value of a **fully rectified** sine wave over a complete cycle is:

$$V_{\text{avg}} = \left(\frac{2}{\pi}\right) V_p = 0.637 V_p, \quad I_{\text{avg}} = \left(\frac{2}{\pi}\right) I_p = 0.637 I_p$$

Formula

Expressed in θ (in degrees or radians)

$$v = V_p \sin \theta \text{ Volt}$$

Expressed with reference to time (time domain expression)

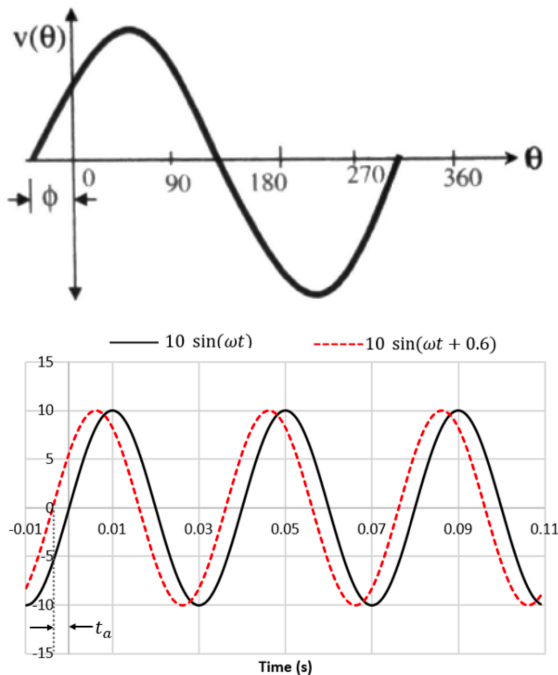
$$v = V_p \sin \omega t = V_p \sin 2\pi f t \text{ Volt}$$

where ω is the angular velocity (in radians/sec) of the rotating coil in the ac generator ($\omega = 2\pi f t$ radians), f is the frequency of the ac sinusoidal voltage produced, and t is for time (in seconds).

Phase-Shift

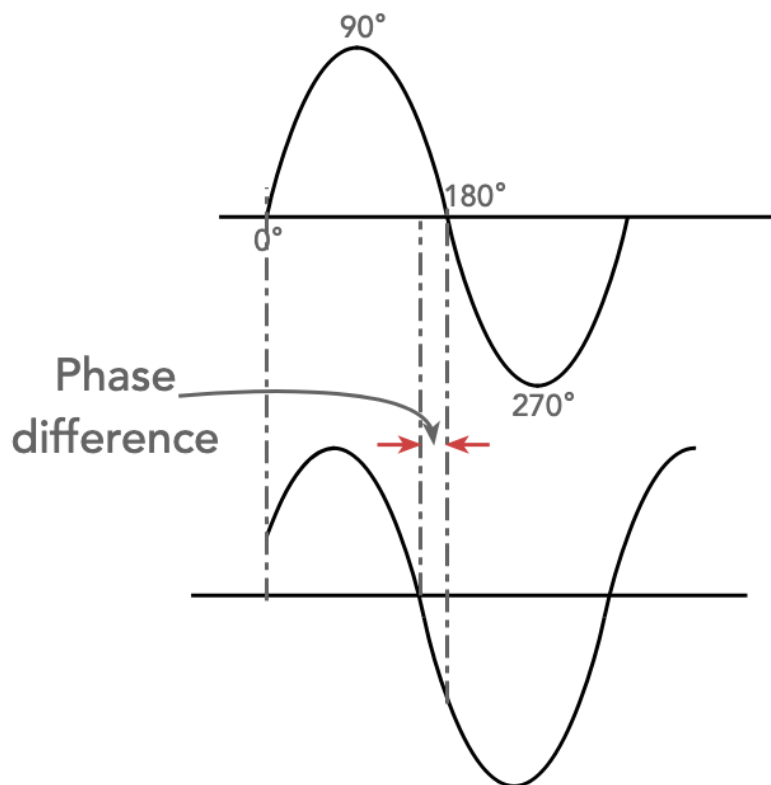
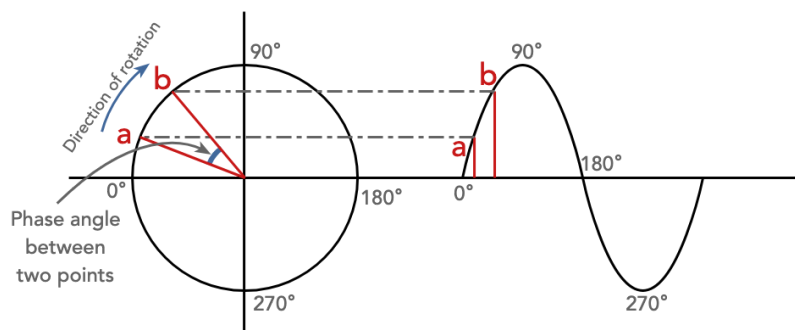
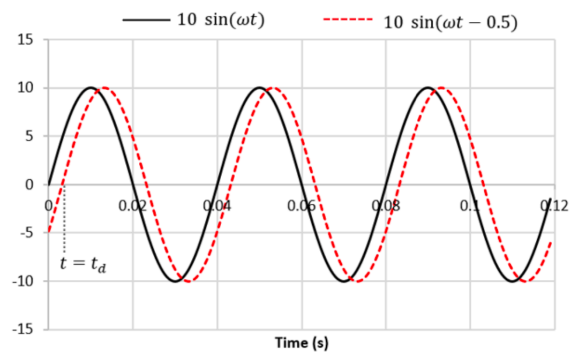
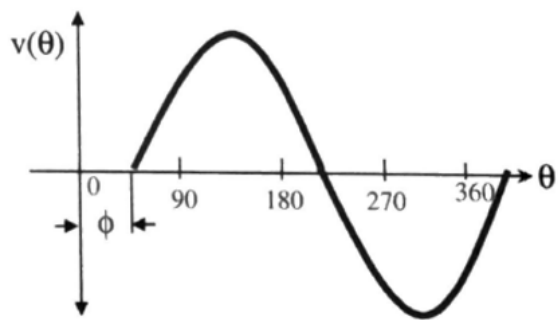
Leading: Sine wave shifted ϕ° to the **left** of the reference means + *phase shift* of ϕ°

$$v = V_p \sin(\theta + \phi) \text{ Volt}$$

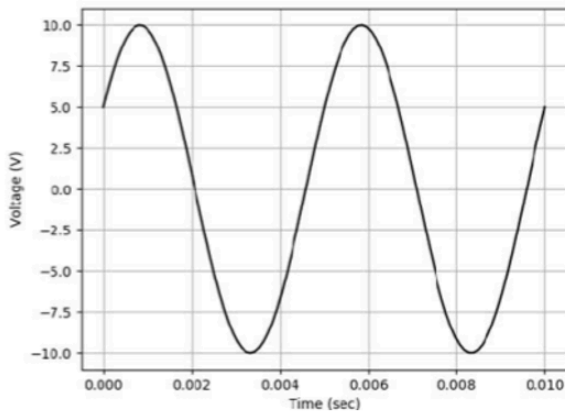


Lagging: Sine wave shifted ϕ° to the **right** of the reference means - *phase shift* of ϕ°

$$v = V_p \sin(\theta - \phi) \text{ Volt}$$



Example :



- Amplitude: 10V
- Period: 0.005s
- frequency: 200Hz
- angular frequency ω : $2 \cdot \pi \cdot f = 2 \cdot \pi \cdot 200 = 1256.6 \text{ rad/s}$
- shift: $10 \sin(0 + \phi) = 5V$, would be 870 which is 30 away from 900 (360+360+180), $\phi : +30^\circ$, or $+\frac{\pi}{6} \text{ rad}$
- $v(t) = 10 \sin(400\pi t + \frac{\pi}{6})$

Oscilloscope

Understanding Coupling Modes

An oscilloscope allows users to choose different coupling modes to analyze signals effectively. The three common modes are:

- **Ground (GND):** Disconnects input and shows a 0V reference baseline.
- **DC Coupling:** Displays both AC and DC components of the signal.
- **AC Coupling:** Blocks the DC component and only shows AC variations.

False Statement in Coupling Mode Question

The **false statement** is:

In DC coupling mode, the average of the waveform is always at 0V.

◆ **Why is this false?**

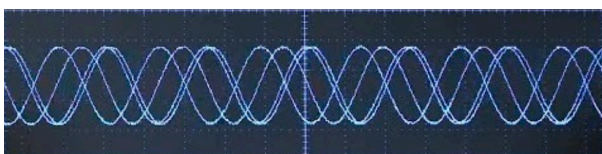
- In **DC coupling mode**, the oscilloscope displays both the **AC and DC** components.
- The **average voltage depends on the DC offset** and is **not necessarily 0V**.

✓ **Explanation of Other Options:**

- ✓ (a) Ground coupling mode shows a horizontal line at 0V.
- ✓ (c) AC coupling blocks the DC average of the signal.
- ✓ (d) In AC coupling mode, the average of the waveform is always at 0V.

Signal Stability Issues on an Oscilloscope

Question: Why does the AC signal appear unstable or moving?



1 Trigger Issues

- **Trigger level not set correctly** → The oscilloscope cannot lock onto a consistent part of the waveform.
- **Trigger channel not set correctly** → If an incorrect channel is selected, the oscilloscope might not detect the expected trigger signal.

2 Incorrect Time Scale

- **Horizontal (Time) Scale Misconfigured** → If set too fast or too slow, the waveform may appear unstable or blurry.
- **Vertical (Voltage) Scale Misconfigured** → A poor vertical scale setting may make the waveform hard to analyze, but it does not cause movement.

Issue	Effect
Incorrect Trigger Level	Causes unstable/moving waveform
Incorrect Trigger Channel	Prevents stable waveform display
Wrong Horizontal Time Scale	Affects waveform clarity
Wrong Vertical Scale	Changes visibility but not stability